Combinatorics 4

Siddharth K

October 2025

Problem 1

Assume $m \leq n$. Then let $B = [m] \times [n] \subseteq [n] \times [n]$. We have that $r_k = 0$ if $k > m$, since at least two rooks would have to be on the same row. Otherwise, we can find an arrangement of rooks by choosing k rows and k columns, and permuting them. This gives us $r_k = k!\binom{m}{k}\binom{n}{k}$, so

$$r_B(x) = \sum_{k=0}^{n} k! \binom{m}{k} \binom{n}{k} x^k.$$

Problem 2

Using Theorem 4.3, we let $t = 10$ since $s_t = c_t - t + 1 = 8 - t + 1 = 9 - t < 0$. We get

$$(s_i) = (0, -1, -2, -3, -3, -4, -5, -5, -4, -1).$$

Then, we have

$$(a_i) = (2, 1, 2, 2, 2, 0, \ldots 0).$$

Plugging into the product, we get

$$\prod_{k \geq 1} \binom{a_k + a_{k+1} - 1}{a_{k+1}} = \binom{2+1-1}{1}\binom{1+2-1}{2}\binom{2+2-1}{2}\binom{2+2-1}{2}\binom{2+0-1}{0}$$

$$= 2 \cdot 1 \cdot 3 \cdot 3 \cdot 1 = 18.$$

So there are 18 Ferrers boards with the same rook polynomial as $(1, 1, 2, 4, 8)$.

Problem 3

We have that $r_B(1) = \sum_{k=0}^{n} r_k$ which is the number of ways to place rooks on B without attacking, or equivalently, the number of ways of choosing k elements of B where no two lie on the same row or column.

Also, $N_B(1) = \sum_{j=0}^{n} N_j$, which is the number of permutations π of $[n]$ where $|B \cap G(\pi)| \in \{0, \ldots, n\}$, or more simply, the number of permutations of $[n]$.

Problem 5

Pick exactly x out of p positions to put an a card and y of q cards to put a b card, with $x+y=k$. Then the remaining $p-x$ a cards must go to the remaining $q-y$ b cards and the remaining $q-y$ b cards must go to the remaining $p-x$ a cards. This forces that

$$p-x=q-y \implies x=\frac{p-q+k}{2}.$$

So the answer is

$$\binom{p}{x}\binom{q}{y}=\binom{p}{\frac{p-q+k}{2}}\binom{q}{\frac{q-p+k}{2}}.$$

Problem 7

For any permutation π, the graph $G(\pi)$ has exactly n points, and every point either lies on B or B^c. So

$$|G(\pi)\cap B^c|=n-|G(\pi)\cap B|.$$

So the numbers satisfy

$$N_{B^c,j}=N_{B,n-j}.$$

Therefore,

$$N_{B^c}(x)=\sum_{j=0}^{n}N_{B^c,j}x^j=\sum_{j=0}^{n}N_{B,n-j}x^j=\sum_{k=0}^{n}N_{B,k}x^k=x^nN_B\left(\frac{1}{x}\right).$$